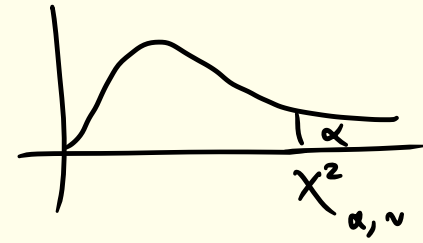


$$f(x) = \frac{1}{2^{n/2} \Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, \quad x > 0 \rightarrow X \sim \chi^2_n$$



F- Dağılımı

Beta için $\propto f$,

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$$

Eğer $\alpha=1$, $\beta=1$ alınırsa standart uniform dağılım,

$$f(x) = 1, \quad 0 < x < 1$$

$$\int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \underbrace{\frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}}_{B(\alpha, \beta)}$$

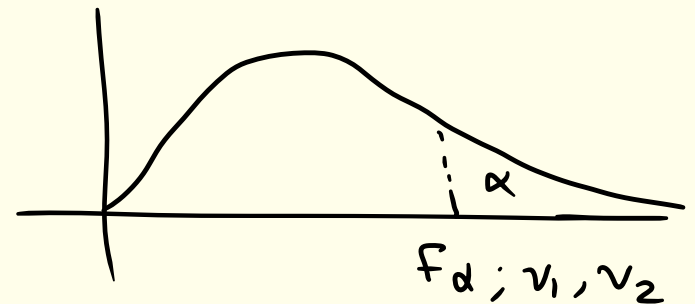
$$\alpha = \frac{\nu_1}{2}, \quad \beta = \frac{\nu_2}{2}$$

$$f(x) = \frac{\Gamma\left(\frac{\nu_1}{2} + \frac{\nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right) \Gamma\left(\frac{\nu_2}{2}\right)} x^{\frac{\nu_1}{2}-1} (1-x)^{\frac{\nu_2}{2}-1}, \quad 0 < x < 1$$

$$y = \frac{\nu_2 x}{\nu_1(1-x)} \quad \text{olsun.}$$

$$f(y) = \underbrace{\frac{\Gamma\left(\frac{\nu_1 + \nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right) \Gamma\left(\frac{\nu_2}{2}\right)}}_{\text{sabit}} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} \underbrace{\frac{y^{\frac{\nu_1}{2}-1}}{\left(1 + \frac{\nu_1}{\nu_2} y\right)^{\frac{\nu_1 + \nu_2}{2}}}}_{\text{değişken}}, \quad y > 0$$

$$\sim F_{\nu_1, \nu_2} \Rightarrow \int_0^{\infty} f(y) dy = \alpha$$



Teorem: $X_1 \sim \chi^2_{\nu_1}$ ve $X_2 \sim \chi^2_{\nu_2}$ bağımsız r.d. leri olsun.

$$Y = \frac{X_1/X_2}{\nu_2/\nu_1} \sim F_{\nu_1, \nu_2} \quad \text{ve} \quad \nu_1 = \nu_2 = 1 \Rightarrow \frac{X_1}{X_2} \sim F_{1,1}$$

Örnek: $N(\mu_1, \sigma_1^2)$ ve $N(\mu_2, \sigma_2^2)$ olsun. $H_0: \sigma_1^2 = \sigma_2^2$
 $H_s: \sigma_1^2 \neq \sigma_2^2$
 $\geq$
 $\leq$

$N(\mu_1, \sigma_1^2) \rightarrow n_1 \rightarrow$ r.ö. örnek varyansı S_1^2

$N(\mu_2, \sigma_2^2) \rightarrow n_2 \rightarrow$ r.ö. örnek varyansı S_2^2

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1}$$

$$\frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}$$

$$\frac{\frac{(\cancel{n_1-1})S_1^2}{\cancel{\sigma_1^2}} / (\cancel{n_1-1})}{\frac{(\cancel{n_2-1})S_2^2}{\cancel{\sigma_2^2}} / (\cancel{n_2-1})} = \frac{S_1^2}{S_2^2} \sim F_{n_1-1; n_2-1}$$

varsayım : $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Teorem : $X \sim F_{\nu_1, \nu_2}$ ise $\frac{1}{X} \sim F_{\nu_2; \nu_1}$ olur.

$$E(X) = \frac{\nu_2}{\nu_2 - 2}$$

$$V(X) = \frac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 4)(\nu_2 - 2)^2}, \quad \nu_2 > 4$$

t-Dağılımı (Student's t-dağılımı)

$N(\mu, \sigma^2)$ dağılımından n bağımsız r.ö. $x_1, x_2, \dots, x_n$ olsun.
Örneklem ort. $\bar{X}$ olarak verilsin.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

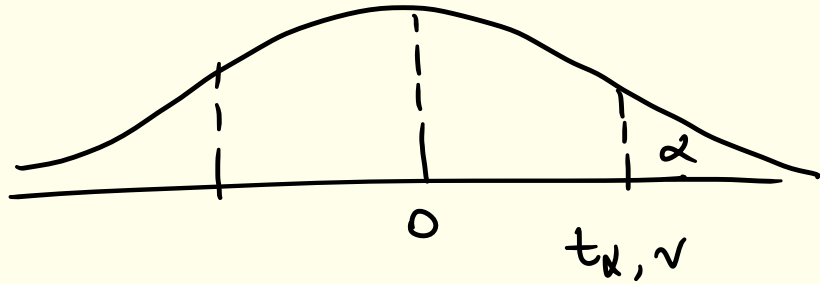
σ^2 bilinmiyorsa, yerine örneklem varyansı S^2 kullanılır.

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t\text{-dağılımı}$$

Genel $X \sim N(0, 1)$, $Y \sim \chi^2_n$ bağımsız.

$$T = \frac{X}{\sqrt{Y/2}} \sim t_n$$

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{2}\right)^{-(\nu+1)/2}, \quad -\infty < t < \infty$$



$$\int_{t_{\alpha, \nu}}^{\infty} f(t) dt = \alpha$$

Teorem: $N(\mu, \sigma^2)$; σ^2 bilinmiyor, n birimlik r.ö.
 $X_1, X_2, \dots, X_n$ olsun.

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1} \text{ olur.}$$

İspat

$$\left. \begin{array}{l} \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1) \\ \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \end{array} \right\} \frac{\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}}{\sqrt{\frac{(n-1)s^2}{\sigma^2} / (n-1)}} = \frac{\bar{X} - \mu}{s / \sqrt{n}} \sim t_{n-1}$$

$\bar{X}$ ve s^2 bağımsız.

$$E(T) = 0 \quad \left. \begin{array}{l} \text{simetrik} \\ \nu(T) = \frac{\nu}{\nu-2} \end{array} \right\}$$

$$E(T) = E\left[\left(\frac{x}{\sqrt{Y/n}}\right)\right] = E(x) E(\sqrt{Y/n}) =$$

$$x \sim N(0,1) \Rightarrow x^2 \sim \chi^2_1, \quad E(x^2) = 1$$

$$V(T) = E(T^2) - [E(T)]^2$$

$$= E\left(\frac{x^2}{y/v}\right) \quad x \text{ ve } y \text{ bağımsız}$$

$$= \underbrace{E(x^2)}_1 \cdot E(v/y)$$

$$= E(v/y) = \int_0^{\infty} \frac{1}{2^{v/2} \Gamma(\frac{v}{2})} y^{\frac{v}{2}-1} e^{-\frac{y}{2}} dy$$

$$= \frac{v}{2^{v/2} \Gamma(\frac{v}{2})} \underbrace{\int_0^{\infty} y^{(\frac{v}{2}-1)-1} e^{-\frac{y}{2}} dy}_{\text{gamma}} = \frac{v}{v-2}$$

→ t-dağılımında n arttıkça std. normal dağılıma dönüşür.

$$\lim_{n \rightarrow \infty} \frac{\nu}{\nu-2} = 1$$

Önemli Sonuç

$$X \sim t_\nu$$

$$X^2 \sim F_{1, \nu}$$

$$\hookrightarrow T = \frac{X}{\sqrt{\gamma/\nu}}$$

$$X \sim N(0,1)$$

$$\gamma \sim \chi^2_\nu$$

bağımsız.

$$\left. \begin{array}{l} X \sim N(0,1) \\ \gamma \sim \chi^2_\nu \end{array} \right\} \frac{X^2}{\gamma/\nu} = \frac{X^2/1}{\gamma/\nu} \sim F_{1, \nu}$$

